

Objectives

The objective of this task is to develop a Python-based temperature conversion application that converts temperatures between Celsius, Fahrenheit, and Kelvin. The project helps practice Python functions, arithmetic operations, conditional statements, input validation, and formatted output while providing a simple interactive interface using Streamlit.

Key Features

- Converts **Celsius to Fahrenheit and Kelvin**.
- Converts **Fahrenheit to Celsius and Kelvin**.
- Converts **Kelvin to Celsius and Fahrenheit**.
- Allows the user to select the source and destination units.
- Accepts decimal temperature values.
- Uses separate functions for each conversion.
- Validates physically meaningful temperature values.
- Displays the converted temperature with two decimal places.
- Displays the formula used for the conversion.
- Provides a clean and user-friendly Streamlit interface.
- Runs locally through the Streamlit development server.

Technologies Used

- **Python**
- **Streamlit**
- Python functions
- Conditional statements
- Arithmetic operators
- Input validation
- HTML/CSS styling through Streamlit

Deliverables

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY14>

```
st.markdown(
    '<div class="subtitle">Convert temperatures between Celsius, Fahrenheit and Kelvin</div>',
    unsafe_allow_html=True
)

def celsius_to_fahrenheit(celsius):
    return (celsius * 9 / 5) + 32

def celsius_to_kelvin(celsius):
    return celsius + 273.15

def fahrenheit_to_celsius(fahrenheit):
    return (fahrenheit - 32) * 5 / 9

def fahrenheit_to_kelvin(fahrenheit):
    return (fahrenheit - 32) * 5 / 9 + 273.15

def kelvin_to_celsius(kelvin):
    return kelvin - 273.15

def kelvin_to_fahrenheit(kelvin):
    return (kelvin - 273.15) * 9 / 5 + 32

col1, col2 = st.columns(2)

with col1:
    from_unit = st.selectbox(
        "Convert From",
        ["Celsius", "Fahrenheit", "Kelvin"]
    )

with col2:
    to_unit = st.selectbox(
        "Convert To",
        ["Celsius", "Fahrenheit", "Kelvin"]
    )

value = st.number_input(
    "Enter Temperature",
    value=0.0,
    step=0.1,
```

Celsius → Fahrenheit: $^{\circ}\text{F} = (^{\circ}\text{C} \times 9/5) + 32$

Celsius → Kelvin: $\text{K} = ^{\circ}\text{C} + 273.15$

Fahrenheit → Celsius: $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$

Fahrenheit → Kelvin: $\text{K} = (^{\circ}\text{F} - 32) \times 5/9 + 273.15$

Kelvin → Celsius: $^{\circ}\text{C} = \text{K} - 273.15$

Kelvin → Fahrenheit: $^{\circ}\text{F} = (\text{K} - 273.15) \times 9/5 + 32$

Github:- <https://github.com/shambhavi0804/VEDA-internship/commit/5ccf680e8b7c060a9a219ec93c1dfa1a74d81293>

```

    DAY14/app.py
    ...
    @@ -0,0 +1,183 @@
    1 + import streamlit as st
    2 + st.set_page_config(
    3 +     page_title="Temperature Converter",
    4 +     page_icon="🌡️",
    5 +     layout="centered"
    6 + )
    7 + st.markdown("""
    8 + <style>
    9 +     .main-title {
    10 +         font-size: 38px;
    11 +         font-weight: 700;
    12 +         text-align: center;
    13 +         margin-bottom: 5px;
    14 +     }
    15 +
    16 +     .subtitle {
    17 +         text-align: center;
    18 +         font-size: 18px;
    19 +         color: #6666;
    20 +         margin-bottom: 30px;
    21 +     }
    22 +
    23 +     .result-box {
    24 +         padding: 20px;
    25 +         border-radius: 12px;
    26 +         text-align: center;
    27 +         margin-top: 20px;
    28 +         border: 1px solid #ddd;
    29 +     }
    30 +
    31 +     .result-value {
    32 +         font-size: 32px;
    33 +         font-weight: 700;

```

Linkedin:- https://lnkd.in/p/dGadw_q2



Temperature Converter

Convert temperatures between Celsius, Fahrenheit and Kelvin

Convert From

Fahrenheit

Convert To

Celsius

Enter temperature

28.75

→

Convert Temperature

Converted Temperature

-1.83 Celsius

Formula:

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$$



Conversion Reference

Celsius → Fahrenheit: $^{\circ}\text{F} = (^{\circ}\text{C} \times 9/5) + 32$

Celsius → Kelvin: $\text{K} = ^{\circ}\text{C} + 273.15$

Fahrenheit → Celsius: $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$

Fahrenheit → Kelvin: $\text{K} = (^{\circ}\text{F} - 32) \times 5/9 + 273.15$

Kelvin → Celsius: $^{\circ}\text{C} = \text{K} - 273.15$

Kelvin → Fahrenheit: $^{\circ}\text{F} = (\text{K} - 273.15) \times 9/5 + 32$